

SQL PROJECT - QUICK-COMMERCE LOGISTICS ANALYTICS

Question Set 1 - Easy

1. Product Category Revenue Analysis

Business Context: Warehouse real estate is expensive. Inventory managers need to understand sales volume distribution across categories to allocate physical shelf space efficiently. High-turnover goods need prime warehouse placement to accelerate the packing process.

- **Task:** Write a query that returns the total sales value and total order count for each product category. Order the results to show the highest-earning category first.

2. Delivery Delay Identification by Platform

Business Context: Delivery delays harm brand reputation. The operations lead needs a quick cross-platform health check to evaluate courier efficiency.

- **Task:** Write a query that returns the name of each platform along with the total count of orders that faced a delivery delay.

3. High-Risk Category for Refunds

Business Context: Refunds represent lost capital, wasted delivery costs, and quality control issues. The Quality Assurance (QA) team needs to pinpoint specific supply lines causing friction.

- **Task:** Write a query to find which product categories trigger the highest volume of refund requests.

4. Customer Service Scorecard

Business Context: Customer experience (CX) and product managers monitor user satisfaction benchmarks relative to their core competition.

- **Task:** Write a query that tracks and returns the average service rating given by users across the different delivery platforms, rounded to two decimal places.

5. Flagging Critical Complaints

Business Context: Severe dissatisfaction paired with financial disputes requires urgent remediation before users permanently churn.

- **Task:** Write a query for the customer support escalation desk that returns the 10 most recent orders where a refund was requested and the customer left a 1-star service rating. Return the Order ID, Platform, Product Category, and Customer Feedback.

Question Set 2 - Moderate

1. Delay Impact on Refund Customer Behavior

Business Context: The business strategy team wants to prove the financial consequences of logistical issues. Specifically, they need to measure how transit failures drive dispute returns.

- **Task:** Write a query that groups orders by their delivery delay status and counts both the total orders and total refund requests to see if delayed orders correlate with higher return rates.

2. High-Value Order Friction Analysis

Business Context: High-ticket transactions account for a significant portion of company revenue. The finance team considers delivery failures on big-budget baskets a high risk.

- **Task:** Write a query that returns the total high-value orders (valued over 1,000 INR) and the count of those premium orders that experienced a delivery delay, grouped by each platform.

3. Customer Satisfaction vs. Speed Thresholds

Business Context: The product team wants to identify the operational "tipping points" where transit speeds impact customer satisfaction.

- **Task:** Write a query that buckets order delivery times into 'Fast (≤ 20 Mins)', 'Slow (> 40 Mins)', and 'Standard (21-40 Mins)'. Return the total order volume and average service rating for each speed classification.

4. Evaluating Lost Capital on Refunds

Business Context: Chief Financial Officers (CFOs) need to audit revenue leakage to adjust quarterly financial reserves.

- **Task:** Write a query to calculate the total amount of capital stuck inside orders where a refund was requested. Break the data down by platform and product category, and order it by the highest value at risk first.